

Roll No. 

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END SEMESTER EXAMINATION - 2024

Name of the Course: **B. Tech Petroleum Engineering**  
Name of the Paper: **Advanced Mathematics**

Semester: **Third**  
Paper Code: **TMA-307**

Time: **3 Hours**

Maximum Marks: **100**

Note: -

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b and c in each main question.
- (iii) Total marks in each main question are twenty.

Each question carries 10 marks.

|     |                                                                                                                                                              |     |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| Q1  | (20 marks)                                                                                                                                                   |     |
| (a) | Derive an expression for complex form of Fourier Integral.                                                                                                   |     |
| (b) | Show using Fourier Integral representation,<br>$\int_0^\infty \left( \frac{\omega \sin x \omega}{1+\omega^2} \right) d\omega = \frac{\pi}{2} e^{-x}, x > 0.$ | CO1 |
| (c) | Find the Fourier sine and cosine transform of $f(x) = e^{-ax}, a > 0$                                                                                        |     |
| Q2  | (20 marks)                                                                                                                                                   |     |
| (a) | State and prove Cauchy fundamental theorem.                                                                                                                  |     |
| (b) | Show that Z transform, $Z(n^p) = -\delta \frac{d}{dz} [z(n^{p-1})], n \geq 0.$                                                                               | CO2 |
| (c) | Find the analytic region of the function $f(x) = (x - y)^2 - 2i(x + y)$                                                                                      |     |
| Q3  | (20 marks)                                                                                                                                                   |     |
| (a) | Expand $f(z) = \cos z$ about $z = \frac{\pi}{2}$ by Taylor's series.                                                                                         | CO3 |
| (b) | Find the image of the region $r \leq 2, 0 \leq \theta \leq \frac{\pi}{4}$ under the map $w = z^2$ .                                                          |     |
| (c) | Evaluate $\int_C \frac{dz}{(z^2+4)^2}$ using Cauchy Residue theorem                                                                                          |     |
| Q4  | (20 marks)                                                                                                                                                   |     |

|     |                                                                                                                                                                                                                                                                                                                                                                                                                        |                                         |       |       |       |       |  |     |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|-------|-------|-------|-------|--|-----|
|     |                                                                                                                                                                                                                                                                                                                                                                                                                        | Compute the mode of the following data: |       |       |       |       |  | CO4 |
| (a) | Expenditure (Rs.)                                                                                                                                                                                                                                                                                                                                                                                                      | 0-10                                    | 10-20 | 20-30 | 30-40 | 40-50 |  |     |
|     | No. of families                                                                                                                                                                                                                                                                                                                                                                                                        | 14                                      | 25    | 27    | 24    | 15    |  |     |
| (b) | (i) A die is thrown twice and the sum of the numbers observed is 7. Find the conditional probability that the number 2 has appeared at least once.<br>(ii) Mukesh rolls a die and Preeti tosses a coin. What is the probability that a Head appears on the coin and a 3 appears on the die?                                                                                                                            |                                         |       |       |       |       |  |     |
| (c) | Suppose the number of tele phone calls on an operator received from 9:00 to 9:05 follow a Poisson's distribution with a mean of 3. Find the probability that the operator will receive at most 2 phone calls in that time interval.                                                                                                                                                                                    |                                         |       |       |       |       |  |     |
| Q5  | (20 marks)                                                                                                                                                                                                                                                                                                                                                                                                             |                                         |       |       |       |       |  | CO5 |
| (a) | Explain the concept of fuzzy sets with membership function. Define the operations and excluded middle axioms on fuzzy sets with diagrams.                                                                                                                                                                                                                                                                              |                                         |       |       |       |       |  |     |
| (b) | A discrete fuzzy set defined on the universe $X = \{m, n, p, q, r, s\}$ is given by $\underline{U} = \{\frac{1}{m}, \frac{0.7}{n}, \frac{0.6}{p}, \frac{0.3}{q}, \frac{0.02}{r}, \frac{0}{s}\}$ .<br>(i) Represent this fuzzy set in a diagram schematically.<br>(ii) List the $\alpha$ -cut sets for the following values of $\alpha$ : 1, 0.7, 0.6, 0.02, 0 and $0^+$ and represent sets in a diagram schematically. |                                         |       |       |       |       |  |     |
| (c) | Relation matrices for the relations R and S are given below. Using max-min composition operation write the composition T for R and S relations.<br>$R = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \text{ and } S = \begin{bmatrix} 0 & 1 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$                                                                                                 |                                         |       |       |       |       |  |     |